

Platform Overload Components

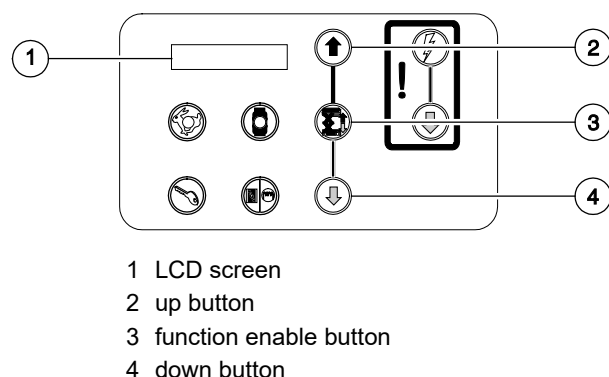
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Platform Overload Recovery Message (software version B3 and later)

If the ground controls LCD screen displays Ovrlld Recovery, the emergency lowering system has been used while the platform was overloaded.

Note: This message shall be cleared by a person trained and qualified on the troubleshooting and repair of this machine.

Note: Use the following illustration to identify the description of each control button used in this procedure.



- 1 Turn the key switch to the ground controls position and pull out the red Emergency Stop button to the on position at the platform controls.
- 2 Press and hold both the blue platform up and yellow platform down buttons. Pull out the red Emergency Stop button to the on position at the ground controls.
- Result: Tune speeds is showing in the LCD screen.

- 3 Use the down button to scroll to select options.
- 4 Press the lift function enable button.
- Result: Descent delay is showing in the LCD screen.
- 5 Use the down button to scroll to overload.
- 6 Press and hold the function enable button for 5 seconds.
- Result: A screen will appear asking for the overload recovery passcode.
- 7 Press the buttons in the following sequence: **(down) (down) (up) (enable)**
- Result: The LCD screen displays cleared and returns to the overload screen.
- 8 Use the down button to scroll to return to main menu.
- 9 Press the lift function enable button.
- Result: Select options is showing in the LCD screen.
- 10 Push in the ground controls red Emergency Stop button to the off position.

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Fault Codes



Observe and Obey:

- ☑ Troubleshooting and repair procedures shall be completed by a person trained and qualified on the repair of this machine
- ☑ Immediately tag and remove from service a damaged or malfunctioning machine.
- ☑ Repair any machine damage or malfunction before operating the machine.
- ☑ Unless otherwise specified, perform each procedure with the machine in the following configuration:
 - Machine parked on a firm, level surface
 - Key switch in the off position with the key removed
 - The red Emergency Stop button in the off position at both the ground and platform controls
 - Wheels chocked
 - All external AC power supply disconnected from the machine
 - Platform in the stowed position

Before Troubleshooting:

- ☑ Read, understand and obey the safety rules and operating instructions in the appropriate operator's manual on your machine.
- ☑ Be sure that all necessary tools and test equipment are available and ready for use.
- ☑ Read each appropriate fault code thoroughly. Attempting short cuts may produce hazardous conditions.
- ☑ Be aware of the following hazards and follow generally accepted safe workshop practices.

⚠ DANGER

Crushing hazard. When testing or replacing any hydraulic component, always support the structure and secure it from movement

⚠ WARNING

Electrocution/burn hazard. Contact with electrically charged circuits could result in death or serious injury. Remove all rings, watches and other jewelry.

⚠ WARNING

Bodily injury hazard. Spraying hydraulic oil can penetrate and burn skin. Loosen hydraulic connections very slowly to allow the oil pressure to dissipate gradually. Do not allow oil to squirt or spray.

- ⦿ Indicates that a specific result is expected after performing a series of steps.
- ⊗ Indicates that an incorrect result has occurred after performing a series of steps.

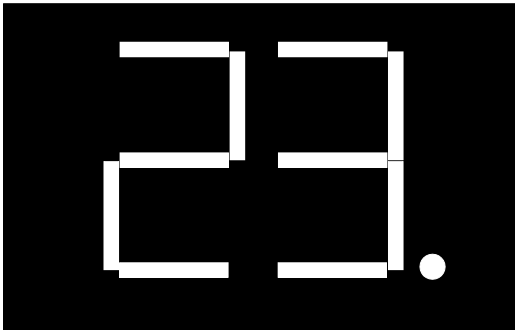
Note: Two persons will be required to safely perform some troubleshooting procedures.

Fault Codes

About This Section

When a malfunction is discovered, the fault code charts in this section will help a service professional pinpoint the cause of the problem. To use this section, basic hand tools and certain pieces of test equipment are required — voltmeter, ohmmeter, pressure gauges.

LED Diagnostic Readout



The diagnostic readout displays numerical codes that provide information about the machine operating status and about malfunctions. The dot to the right of the numbers remain on when a fault code is displayed.

The codes listed in the Fault Code Chart describe malfunctions and can aid in troubleshooting the machine by pinpointing the area or component affected.

Fault Codes

ID	System Ready	LED	Condition	Result	Solution
1	INTERNAL ECU FAULT	Red	Internal ECM error.	System shutdown.	Replace ECM.
2	PLATFORM ECU FAULT	Red	Platform/ECM communication error.	System shutdown.	Troubleshoot control cable OR troubleshoot platform controls.
20	CHASSIS START SW FAULT	Red	Engine start button fault at ground controls.	Engine will not start.	Replace ECM.
21	CHASSIS CHOKE SW FAULT	Red	Starting aid button fault at ground controls.	Starting aid disabled.	Replace ECM.
22	CHASSIS UP SW FAULT	Red	Up switch fault at ground controls.	Platform up function inhibited.	Replace ECM.
23	CHASSIS LIFT SW FAULT	Red	Platform up/down enable button fault at ground controls.	Platform up/down functions inhibited.	Replace ECM.
24	DOWN SW FAULT	Red	Down switch fault at ground controls.	Platform down function inhibited.	Replace ECM.
25	LEFT TURN SW FAULT	Red	Left turn switch fault.	Malfunctioning steer left microswitch.	Troubleshoot steer left microswitch.
26	RIGHT TURN SW FAULT	Red	Right turn switch fault.	Malfunctioning steer right microswitch.	Troubleshoot steer right microswitch.
27	DRIVE ENABLE SW FLT	Red	Function enable switch on joystick is activated when machine is turned on.	Machine functions disabled.	Release function enable switch on joystick before power up OR replace joystick.
28	OFF NEUTRAL DRIVE JOYSTICK	Red	Drive joystick off neutral when machine is turned on.	Machine functions disabled.	Release joystick before power up OR replace joystick.
29	PLATFORM LIFT SW FAULT	Red	Lift enable button fault at the platform controls.	Platform up/down functions inhibited.	Troubleshoot button OR replace printed circuit board at platform controls.
30	OFF NEUTRAL LIFT JOYSTICK	Red	Up/down switch off neutral.	Platform up/down functions inhibited.	Replace up/down switch at platform controls.
31	PLATFORM CHOKE SW FAULT	Red	Starting aid fault at the platform controls.	Starting aid disabled.	Troubleshoot button OR replace printed circuit board at platform controls.

Fault Codes

ID	System Ready	LED	Condition	Result	Solution
32	PLATFORM START SW FAULT	Red	Engine start button fault at platform controls.	Engine will not start.	Troubleshoot button OR replace printed circuit board at platform controls.
33	LEFT FRONT OUTRIG SW FLT	Red	Left front outrigger enable button fault at platform controls.	Outriggers disabled.	Troubleshoot button OR replace printed circuit board at platform controls.
34	RIGHT FRONT OUTRIG SW FLT	Red	Right front outrigger enable button fault at platform controls.	Outriggers disabled.	Troubleshoot button OR replace printed circuit board at platform controls.
35	LEFT REAR OUTRIG SW FLT	Red	Left rear outrigger enable button fault at platform controls.	Outriggers disabled.	Troubleshoot button OR replace printed circuit board at platform controls.
36	RIGHT REAR OUTRIG SW FLT	Red	Right rear outrigger enable button fault at platform controls.	Outriggers disabled.	Troubleshoot button OR replace printed circuit board at platform controls.
37	AUTO LEVEL SWITCH FAULT	Red	Outrigger auto level enable button fault at platform controls.	Outriggers disabled.	Troubleshoot button OR replace printed circuit board at platform controls.
3812	OSCILLATE LIMIT SWITCHES	Red	Value to high. Voltage above 3000 mV.	Elevated drive and platform up disabled.	Troubleshoot oscillate limit switches.
3815	OSCILLATE LIMIT SWITCHES	Red	Value to low. Voltage below 1000 mV.	Elevated drive and platform up disabled.	Troubleshoot oscillate limit switches.
3819	OSCILLATE LIMIT SWITCHES	Red	Voltage out of range.	Elevated drive and platform up disabled.	Troubleshoot oscillate limit switches.
3821	OSCILLATE LIMIT SWITCHES	Red	Voltage inconsistent.	Elevated drive and platform up disabled.	Troubleshoot oscillate limit switches.
4021	OSCILLATE OUTPUT	Red	Both oscillate valve outputs are on at the same time while elevated. Flashing or Motion Beacon option is set to on and it is not installed on the machine.	Elevated drive and platform up disabled.	Troubleshoot oscillate limit switches or replace ECU. Set the option to off if not installed on the machine.
4022	OSCILLATE OUTPUT	Red	Oscillate valve output is active for more than 4 seconds.	Elevated drive and platform up disabled.	Troubleshoot oscillate limit switches or replace ECU.
49	DRIVE COIL 1 FAULT	Red	Drive coil 1 (Y1) fault.	High speed drive function will not operate.	Troubleshoot coil OR wiring.

Fault Codes

ID	System Ready	LED	Condition	Result	Solution
50	DRIVE COIL 2 FAULT	Red	Drive coil 2 (Y1A) fault.	High speed drive function will not operate.	Troubleshoot coil OR wiring.
51	DRIVE COIL 3 FAULT	Red	Drive coil 3 (Y1B) fault.	High speed drive function will not operate.	Troubleshoot coil OR wiring.
52	FUNC PROP COIL FAULT	Red	Proportional coil (Y9) fault.	Lift and outrigger functions are disabled.	Troubleshoot coil OR wiring.
54	UP COIL FAULT	Red	Up coil (Y8) fault.	Platform up function inhibited.	Troubleshoot coil OR wiring.
55	DOWN COIL FAULT	Red	Down coil (Y7) fault.	Platform down function inhibited.	Troubleshoot coil OR wiring.
56	RIGHT TURN COIL FAULT	Red	Right turn coil (Y3) fault.	Steer right function inhibited.	Troubleshoot coil OR wiring.
57	LEFT TURN COIL FAULT	Red	Left turn coil (Y4) fault.	Steer left function inhibited.	Troubleshoot coil OR wiring.
58	BRAKE COIL FAULT	Red	Brake release coil (Y2) fault.	Brakes will not release.	Troubleshoot coil OR wiring.
60	FORWARD 1 COIL FAULT	Red	Forward 1 coil (Y6) fault.	Drives slow or not at all when elevated.	Troubleshoot coil OR wiring.
61	REVERSE 1 COIL FAULT	Red	Reverse 1 coil (Y5) fault.	Drives slow or not at all when elevated.	Troubleshoot coil OR wiring.
62	FORWARD 2 COIL FAULT	Red	Forward 2 coil (Y6A) fault.	Drives slow or not at all when elevated.	Troubleshoot coil OR wiring.
63	REVERSE 2 COIL FAULT	Red	Reverse 2 coil (Y5A) fault.	Drives slow or not at all when elevated.	Troubleshoot coil OR wiring.
64	OSCILLATE LEFT COIL	Red	Value to low or Value to high.	Elevated drive and platform up disabled.	Troubleshoot oscillate left valve coil (Y93) or wiring.
65	OSCILLATE RIGHT COIL	Red	Value to low or Value to high.	Elevated drive and platform up disabled.	Troubleshoot oscillate right valve coil (Y94) or wiring.
66	LOW OIL PRESSURE	Red	Low oil pressure.	Engine stops.	Check the engine oil level OR check wiring from the oil pressure switch to ECM OR replace the oil pressure switch.

Fault Codes

ID	System Ready	LED	Condition	Result	Solution
67	HIGH COOLANT TEMPERATURE	Red	High coolant temperature.	High engine temperature OR defective engine coolant or oil temperature switch.	Gasoline/LPG models: Check the engine radiator coolant level OR check the wiring from the water temperature switch to ECM OR replace the water temperature switch. Diesel models: Check the engine oil level OR check the wiring from the oil temperature switch to ECM OR replace the oil temperature switch.
68	LOW ECU VOLTAGE	Red	Low ECM voltage.	System shutdown.	Charge battery.
69	LOW ENGINE RPM	Red	Low RPM.	Engine idle RPM too low.	Contact Genie Product Support.
70	HIGH ENGINE RPM	Red	High RPM.	Engine RPM too high.	Contact Genie Product Support.
80	LEFT FRONT OTRG COIL FLT	Red	Left front outrigger coil (Y35) fault.	Left front outrigger function inhibited.	Troubleshoot coil OR wiring.
81	LEFT REAR OTRG COIL FLT	Red	Left rear outrigger coil (Y33) fault.	Left front outrigger function inhibited.	Troubleshoot coil OR wiring.
82	RIGHT FRONT OTRG COIL FLT	Red	Right front outrigger coil (Y36) fault.	Right front outrigger function inhibited.	Troubleshoot coil OR wiring.
83	RIGHT REAR OTRG COIL FLT	Red	Right rear outrigger coil (Y34) fault.	Right rear outrigger function inhibited.	Troubleshoot coil OR wiring.
84	OUTRIGGER EXT COIL FLT	Red	Outrigger extend coil (Y40) fault.	Outrigger extend function disabled.	Troubleshoot coil OR wiring.
85	OUTRIGGER RET COIL FLT	Red	Outrigger retract coil (Y39) fault.	Outrigger retract function disabled.	Troubleshoot coil OR wiring.
86	OUTRIGGER SLOW COIL FLT	Red	Outrigger slowdown coil (Y44) fault.	Outrigger slow extend function disabled.	Troubleshoot coil OR wiring.
90	2 SPEED COIL FAULT	Red	2 speed coil (Y1) fault	High torque drive function disabled.	Troubleshoot coil OR wiring.
92	DRIVE FWD PROP COIL FAULT	Red	Drive pump forward proportional coil (Y51) fault	Drive forward function disabled.	Troubleshoot coil OR wiring.
93	DRIVE REV PROP COIL FAULT	Red	Drive pump reverse proportional coil (Y51) fault	Drive reverse function disabled.	Troubleshoot coil OR wiring.
94	MACHINE TYPE FAULT	Red	Wrong machine type selected	Machine functions disabled.	Correct selection.
102	DOWN LS CROSS-CHECK FLT	Red	Down limit switch fault	All functions inhibited except platform down.	Troubleshoot down limit switches and wiring.

Fault Code Display - Deutz Models

How to Retrieve Active Engine Fault Codes - Deutz D 2.9 L4 Models

The ECM constantly monitors the engine by the use of sensors on the engine. The ECM also uses signals from the sensors to initiate sequential fuel injection and make constant and instantaneous changes to ignition timing, fuel delivery and throttle position to maintain the engine's running condition at its highest efficiency while at the same time keeping exhaust emissions to a minimum. When a sensor fails or returns signals that are outside of set parameters, the ECM will store a fault code in memory that relates to the appropriate sensor. One or more fault LED's will illuminate on the display located at the ground control box. The active fault code will also be displayed on the LCD screen.

If an engine fault occurs that does not result in an engine shutdown, the engine rpm will go into limp home mode resulting in the loss of high rpm.

When operating from the platform, if the red Emergency Stop button is pushed in, the active fault code(s) will be erased from the display.

Start the engine from the ground control box and operate various boom functions to verify that an active engine fault occurs and is shown on the display.

Note: All faults are stored in the Previous Fault history menu. These faults will not be erased when corrective action has been completed.

With an active fault and the engine running: (preferred method)

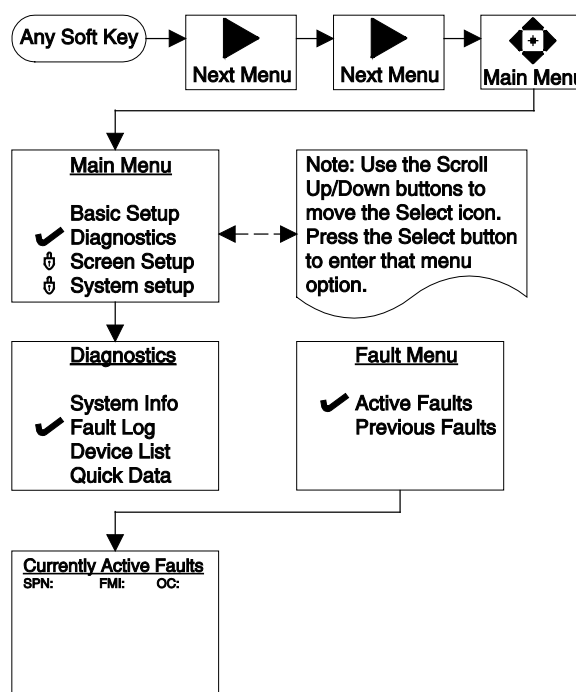
- 1 At the ground controls, activate the auxiliary pump toggle switch to shut the engine off.

Note: Do not push in the red Emergency Stop button or turn the key switch to the off position.

- 2 Press any soft key below the display.
- 3 Use the scroll up / down keys to check for multiple engine fault codes.

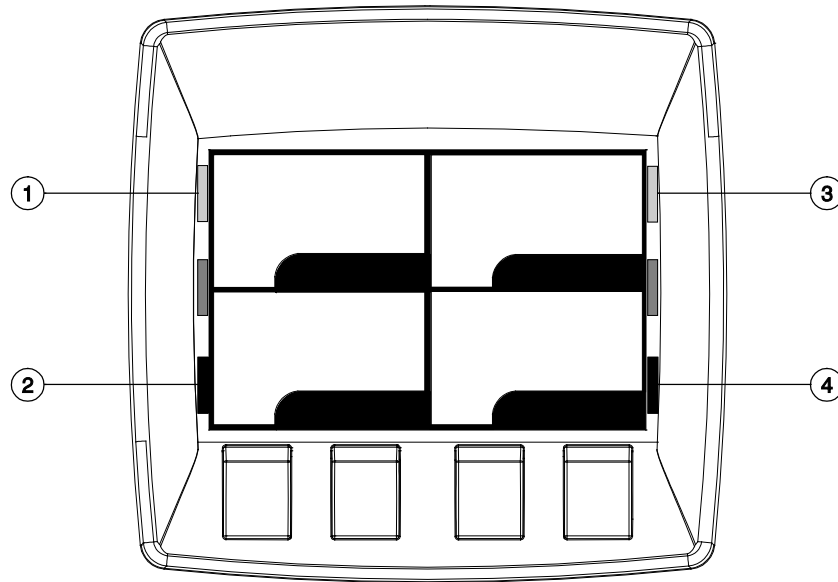
With the engine not running:

- 1 At the ground controls, turn the key switch to ground controls and pull out the red Emergency Stop button.
- 2 Navigate to the Active Fault Menu and use the scroll up / down keys to check for multiple engine fault codes.



Fault Code Display - Deutz Models

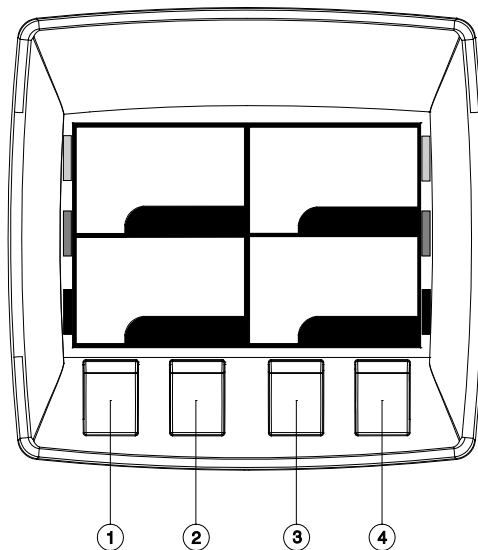
Flashing and Solid LED's - Deutz D 2.9 L4 Models



- 1 Left green LED:
 - Flashing, engine fault detected. Contact service.
 - Solid, fault acknowledged. Contact service.
- 2 Left red LED:
 - Flashing, engine fault detected. Contact service.
 - Flashing with right flashing amber LED, engine soot level over 140%. Engine shut down. Contact service.
- 3 Right green LED:
 - Flashing, engine fault detected. Contact service.
 - Solid, fault acknowledged. Contact service.
- 4 Right red LED:
 - Flashing, engine fault detected. Contact service.
 - Flashing with right flashing amber LED, engine soot level over 140%. Engine shut down. Contact service.

Fault Code Display - Deutz Models

Soft Key Functions and Icons - Deutz D 2.9 L4 Models

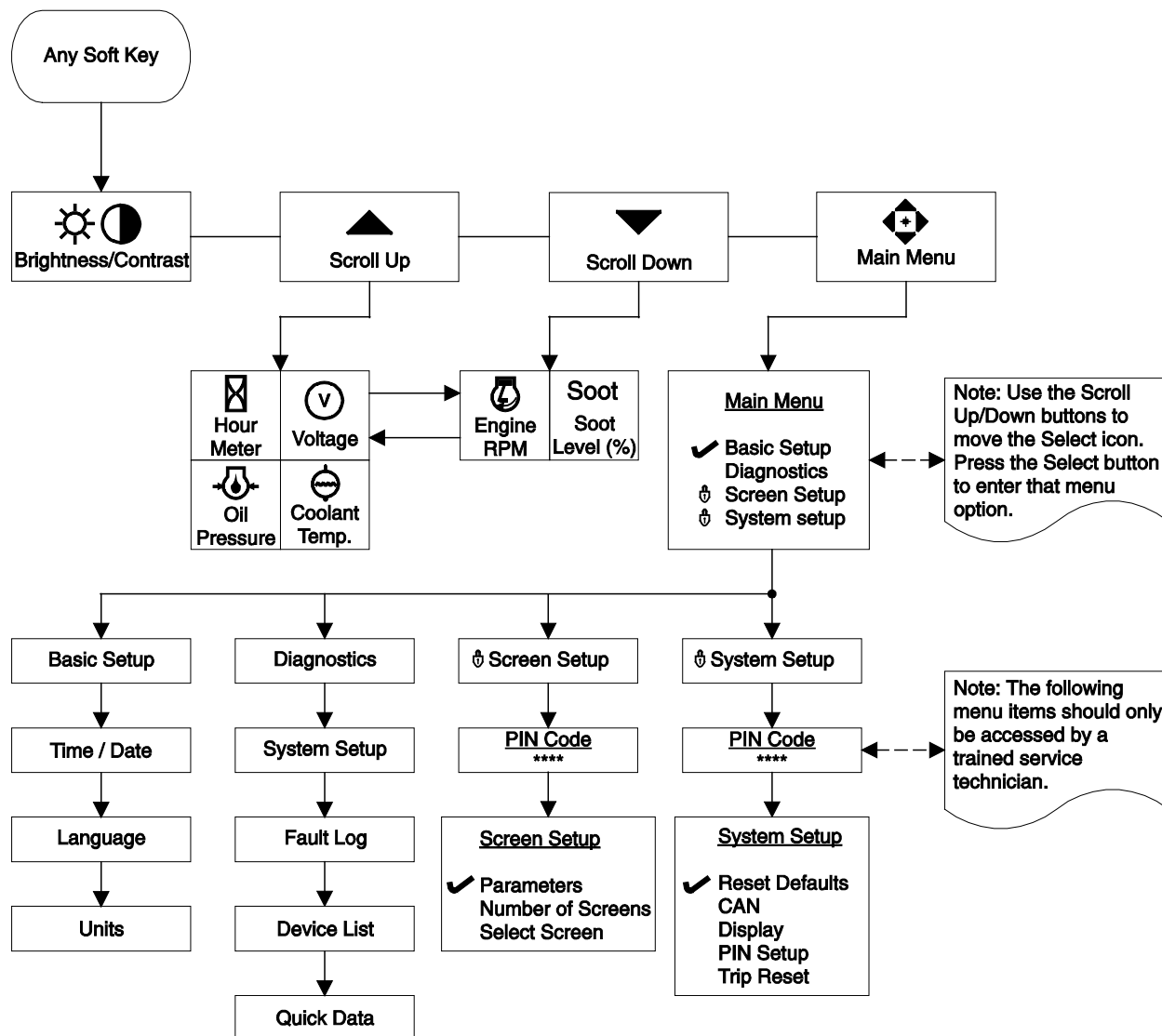


- 1 Next menu • Exit / Back one screen • Decrease brightness (-)
- 2 Brightness / Contrast • Scroll up • Increase • Increase brightness (+)
- 3 Scroll down • Decrease • Decrease contrast (-)
- 4 Select / Next • Main menu • Increase contrast (+)

 Next Menu	 Brightness/ Contrast		
 Exit / Back One Screen	 Scroll Up	 Scroll Down	 Select
 Main Menu	 Hour Meter	 Oil Pressure	 Coolant Temp.
 Engine RPM	 Soot Soot Level Percent (%)	 Voltage	 Pin # Protected

Fault Code Display - Deutz Models

Main Menu Structure - Deutz D 2.9 L4 Models



Deutz D 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
51	3	1019	EGR-Valve, short circuit to battery
51	3	1024	Position sensor error of actuator EGR-Valve; signal range check high
51	3	1226	EGR-Valve; short circuit to battery (A02)
51	3	1227	EGR-Valve; short circuit to battery (A67)
51	4	1020	EGR-Valve; short circuit to ground
51	4	1025	Position sensor error actuator EGR-Valve; signal range check low
51	4	1228	EGR-Valve; short circuit to ground (A02)
51	4	1229	EGR-Valve; short circuit to ground (A67)
1	4	1232	Actuator error EGR-Valve; Voltage below threshold
51	5	1015	Actuator error EGR-Valve; signal range check low
51	5	1017	Actuator EGR-Valve; open load
51	5	1023	Actuator error EGR-Valve; signal range check low
51	5	1223	Actuator EGR-Valve; open load
51	6	1014	Actuator error EGR-Valve; signal range check high
51	6	1022	Actuator error EGR-Valve; signal range check high
51	6	1224	Actuator EGR-Valve; over current
51	6	1230	Actuator error EGR-Valve; Overload by short-circuit
51	7	1016	Actuator position for EGR-Valve not plausible

SPN	FMI	KWP	Description
51	11	1231	Actuator error EGR-Valve; Power stage over temp due to high current
51	12	1018	Actuator EGR-Valve; powerstage over temperature
51	12	1021	Mechanical actuator defect EGR-Valve
51	12	1225	Actuator EGR-Valve; over temperature
94	1	474	Low fuel pressure; warning threshold exceeded
94	1	475	Low fuel pressure; shut off threshold exceeded
94	3	472	Sensor error low fuel pressure; signal range check high
94	4	473	Sensor error low fuel pressure; signal range check low
97	3	464	Sensor error water in fuel; signal range check high
97	4	465	Sensor error water in fuel; signal range check low
97	12	1157	Water in fuel level prefilter; maximum value exceeded
100	0	734	High oil pressure; warning threshold exceeded
100	0	735	High oil pressure; shut off threshold exceeded
100	1	736	Low oil pressure; warning threshold exceeded
100	1	737	Low oil pressure; shut off threshold exceeded
100	3	732	Sensor error oil pressure; signal range check high
100	4	733	Sensor error oil pressure sensor; signal range check low
102	2	88	Charged air pressure above warning threshold
102	2	89	Charged air pressure above shut off threshold
102	4	777	Sensor error charged air press.; signal range check low

Deutz D 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
105	0	996	High charged air cooler temperature; warning threshold exceeded
105	0	997	High charged air cooler temperature; shut off threshold exceeded
105	3	994	Sensor error charged air temperature; signal range check high
105	4	995	Sensor error charged air temperature; signal range check low
108	3	412	Sensor error ambient air press.; signal range check high
108	4	413	Sensor error ambient air press.; signal range check low
110	0	98	High coolant temperature; warning threshold exceeded
110	0	99	High coolant temperature; shut off threshold exceeded
110	3	96	Sensor error coolant temp.; signal range check high
110	4	97	Sensor error coolant temp.; signal range check low
111	1	101	Coolant level too low
132	11	1	Air flow sensor load correction factor exceeding the maximum drift limit; plausibility error
132	11	2	Air flow sensor load correction factor exceeding drift limit; plausibility error
132	11	3	Air flow sensor low idle correction factor exceeding the maximum drift limit
132	11	4	Air flow sensor load correction factor exceeding the maximum drift limit
157	3	877	Sensor error rail pressure; signal range check high
157	4	878	Sensor error rail pressure; signal range check low

SPN	FMI	KWP	Description
168	0	1180	Physical range check high for battery voltage
168	1	1181	Physical range check low for battery voltage
168	2	47	High battery voltage; warning threshold exceeded
168	2	48	Low battery voltage; warning threshold exceeded
168	3	45	Sensor error battery voltage; signal range check high
168	4	46	Sensor error battery voltage; signal range check low
171	3	417	Sensor error environment temperature; signal range check high
171	4	418	Sensor error environment temperature; signal range check low
172	0	1182	Physical range check high for intake air temperature
172	1	1183	Physical range check low for intake air temperature
172	2	9	Sensor ambient air temperature; plausibility error
172	2	983	Intake air sensor; plausibility error
172	3	981	Sensor error intake air; signal range check high
172	4	982	Sensor error intake air sensor; signal range check low
174	0	481	High low fuel temperature; warning threshold exceeded
174	0	482	High Low fuel temperature; shut off threshold exceeded
175	0	740	Physical range check high for oil temperature
175	0	745	High oil temperature; warning threshold exceeded
175	0	746	High oil temperature; shut off threshold exceeded
175	1	741	Physical range check low for oil temperature

Deutz D 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

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SPN	FMI	KWP	Description
175	2	738	Sensor oil temperature; plausibility error
175	2	739	Sensor oil temperature; plausibility error oil temperature too high
175	3	743	Sensor error oil temperature; signal range check high
175	4	744	Sensor error oil temperature; signal range check low
190	0	389	Engine speed above warning threshold (FOC-Level 1)
190	2	421	Offset angle between crank- and camshaft sensor is too large
190	8	419	Sensor camshaft speed; disturbed signal
190	8	422	Sensor crankshaft speed; disturbed signal
190	11	390	Engine speed above warning threshold (FOC-Level 2)
190	12	420	Sensor camshaft speed; no signal
190	12	423	Sensor crankshaft speed; no signal
190	14	391	Engine speed above warning threshold (Overrun Mode)
190	14	1222	Camshaft- and Crankshaft speed sensor signal not available on CAN
411	0	791	Physical range check high for differential pressure Venturiunit (EGR)
411	1	792	Physical range check low for differential pressure Venturiunit (EGR)
411	3	795	Sensor error differential pressure Venturiunit (EGR); signal range check high
411	4	381	Physical range check low for EGR differential pressure
411	4	796	Sensor error differential pressure Venturiunit (EGR); signal range check low

SPN	FMI	KWP	Description
412	3	1007	Sensor error EGR cooler downstream temperature; signal range check high
412	4	1008	Sensor error EGR cooler downstream temperature; signal range check low
520	9	306	Timeout Error of CAN-Receive-Frame TSC1TR; Setpoint
597	2	49	Break lever mainswitch and break lever redundancy switch status not plausible
624	3	971	SVS lamp; short circuit to batt.
624	4	972	SVS lamp; short circuit to grd.
624	5	969	SVS lamp; open load
624	12	970	SVS lamp; powerstage over temperature
630	12	376	Access error EEPROM memory (delete)
630	12	377	Access error EEPROM memory (read)
630	12	378	Access error EEPROM memory (write)
639	14	84	CAN-Bus 0 "BusOff-Status"
651	3	580	Injector 1 (in firing order); short circuit
651	4	586	High side to low side short circuit in the injector 1 (in firing order)
651	5	568	Injector 1 (in firing order); interruption of electric connection
652	3	581	Injector 2 (in firing order); short circuit
652	4	587	High side to low side short circuit in the injector 2 (in firing order)
652	5	569	Injector 2 (in firing order); interruption of electric connection
653	3	582	Injector 3 (in firing order); short circuit
653	4	588	High side to low side short circuit in the injector 3 (in firing order)
653	5	570	Injector 3 (in firing order); interruption of electric connection

Deutz D 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

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KWP = Keyword Protocol

SPN	FMI	KWP	Description
654	3	583	Injector 4 (in firing order); short circuit
654	4	589	High side to low side short circuit in the injector 4 (in firing order)
654	5	571	Injector 4 (in firing order); interruption of electric connection
676	11	543	Cold start aid relay error.
676	11	544	Cold start aid relay open load
677	3	956	Starter relay high side; short circuit to battery
677	3	960	Starter relay low side; short circuit to battery
677	4	957	Starter relay high side; short circuit to ground
677	4	961	Starter relay low side; short circuit to ground
677	5	958	Starter relay; no load error
677	12	959	Starter relay; powerstage over temperature
703	3	426	Engine running lamp; short circuit to battery
703	4	427	Engine running lamp; short circuit to ground
703	5	424	Engine running lamp; open load
703	12	425	Engine running lamp; powerstage over temperature
729	5	545	Cold start aid relay open load
729	12	547	Cold start aid relay; over temperature error
898	9	305	Timeout Error of CAN-Receive-Frame TSC1TE; Setpoint
1079	13	946	Sensor supply voltage monitor 1 error (ECU)
1080	13	947	Sensor supply voltage monitor 2 error (ECU)

SPN	FMI	KWP	Description
1109	2	121	Engine shut off demand ignored
1136	0	1398	Physikal range check high for ECU temperature
1136	1	1399	Physikal range check low for ECU temperature
1136	3	1400	Sensor error ECU temperature; signal range check high
1136	4	1401	Sensor error ECU temperature; signal range check low
1176	3	849	Sensor error pressure sensor upstream turbine; signal range check high
1176	4	850	Sensor error pressure sensor downstream turbine; signal range check high
1180	0	1193	Physical range check high for exhaust gas temperature upstream turbine
1180	0	1460	Turbocharger Wastegate CAN feedback; warning threshold exceeded
1180	0	1462	Exhaust gas temperature upstream turbine; warning threshold exceeded
1180	1	1194	Physical range check low for exhaust gas temperature upstream turbine
1180	1	1461	Turbocharger Wastegate CAN feedback; shut off threshold exceeded
1180	1	1463	Exhaust gas temperature upstream turbine; shut off threshold exceeded
1180	3	1067	Sensor error exhaust gas temperature upstream turbine; signal range check high
1180	11	1066	Sensor exhaust gas temperature upstream turbine; plausibility error
1188	2	1414	Wastegate; status message from ECU missing

Deutz D 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

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SPN	FMI	KWP	Description
1188	7	1415	Wastegate actuator; blocked
1188	11	1411	Wastegate actuator; internal error
1188	11	1412	Wastegate actuator; EOL calibration not performed correctly
1188	11	1416	Wastegate actuator; over temperature (> 145°C)
1188	11	1417	Wastegate actuator; over temperature (> 135°C)
1188	11	1418	Wastegate actuator; operating voltage error
1188	13	1413	Wastegate actuator calibration deviation too large, recalibration required
1231	14	85	CAN-Bus 1 "BusOff-Status"
1235	14	86	CAN-Bus 2 "BusOff-Status"
1237	2	747	Override switch; plausibility error
1322	12	610	Too many recognized misfires in more than one cylinder
1323	12	604	Too many recognized misfires in cylinder 1 (in firing order)
1324	12	605	Too many recognized misfires in cylinder 2 (in firing order)
1325	12	606	Too many recognized misfires in cylinder 3 (in firing order)
1326	12	607	Too many recognized misfires in cylinder 4 (in firing order)
2659	0	1524	Physical range check high for EGR exhaust gas mass flow
2659	1	1525	Physical range check low for EGR exhaust gas mass flow
2659	2	1523	Exhaust gas recirculation AGS sensor; plausibility error

SPN	FMI	KWP	Description
2659	2	1527	AGS sensor temperature exhaust gas mass flow; plausibility error
2659	12	1526	Exhaust gas recirculation; AGS sensor has "burn off" not performed
2797	4	1337	Injector diagnostics; timeout error of short circuit to ground measurement cyl. Bank 0
2798	4	1338	Injector diagnostics; timeout error of short circuit to ground measurement cyl. Bank 1
2798	4	1339	Injector diagnostics; short circuit to ground monitoring Test in Cyl. Bank 0
2798	4	1340	Injector diagnostics; short circuit to ground monitoring Test in Cyl. Bank 1
3224	2	127	DLC Error of CAN-Receive-Frame AT1IG1 NOX Sensor (SCR-system upstream cat; DPF-system downstream cat); length of frame incorrect
3224	9	128	Timeout Error of CAN-Receive-Frame AT1IG1; NOX sensor upstream
3248	4	1047	Sensor error particle filter downstream temperature; signal range check low
3699	2	1616	DPF differential pressure sensor and a further sensor or actuator CRT system defective
3699	2	1617	Temperature sensor us. and ds. DOC simultaneously defect
3699	14	1615	Maximum stand-still-duration reached; oil exchange required
4765	0	1039	Physical range check high for exhaust gas temperature upstream (DOC)
4765	1	1042	Physical range check low for exhaust gas temperature upstream (DOC)

Deutz D 2.9 L4 Engine Fault Codes

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SPN	FMI	KWP	Description
4766	0	1029	Physical range check high for exhaust gas temperature downstream (DOC)
4766	1	1032	Physical range check low for exhaust gas temperature downstream (DOC)
4768	2	1036	Sensor exhaust gas temperature upstream (DOC); plausibility error
4768	3	1044	Sensor error exhaust gas temperature upstream (DOC); signal range check high
4768	4	1045	Sensor error exhaust gas temperature upstream (DOC) signal range check low
4769	2	1026	Sensor exhaust gas temperature downstream (DOC); plausibility error
4769	3	1034	Sensor error exhaust gas temperature downstream (DOC); signal range check high
4769	4	1035	Sensor error exhaust gas temperature downstream (DOC); signal range check low
523006	3	34	Controller mode switch; short circuit to battery
523006	4	35	Controller mode switch; short circuit to ground
523008	1	648	Manipulation control was triggered
523008	2	649	Timeout error in Manipulation control
523009	9	825	Pressure Relief Valve (PRV) reached maximum allowed opening count
523009	10	833	Pressure relief valve (PRV) reached maximum allowed open time

SPN	FMI	KWP	Description
523212	9	171	Timeout Error of CAN-Receive-Frame ComEngPrt; Engine Protection
523216	9	198	Timeout Error of CAN-Receive-Frame PrHtEnCmd; pre-heat command, engine command
523240	9	179	Timeout CAN-message FunModCtl; Function Mode Control
523350	4	565	Injector cylinder-bank 1; short circuit
523352	4	566	Injector cylinder-bank 2; short circuit
523354	12	567	Injector powerstage output defect
523470	2	826	Pressure Relief Valve (PRV) forced to open; performed by pressure increase
523470	2	827	Pressure Relief Valve (PRV) forced to open; performed by pressure shock
523470	7	876	Maximum rail pressure in limp home mode exceeded (PRV)
523470	11	831	The PRV can not be opened at this operating point with a pressure shock
523470	11	832	Rail pressure out of tolerance range
523470	12	828	Open Pressure Relief Valve (PRV); shut off condition
523470	12	829	Open Pressure Relief Valve (PRV); warning condition
523470	14	830	Pressure Relief Valve (PRV) is open
523550	12	980	T50 start switch active for too long
523601	13	948	Sensor supply voltage monitor 3 error (ECU)
523603	9	126	Timeout Error of CAN-Receive-Frame AMB; Ambient Temperature Sensor

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SPN	FMI	KWP	Description
523605	9	300	Timeout Error of CAN-Receive-Frame TSC1AE; Traction Control
523606	9	301	Timeout Error of CAN-Receive-Frame TSC1AR; Retarder
523612	12	387	Internal software error ECU; injection cut off
523612	12	612	Internal ECU monitoring detection reported error
523612	12	613	Internal ECU monitoring detection reported error
523612	12	614	Internal ECU monitoring detection reported error
523612	12	615	Internal ECU monitoring detection reported error
523612	12	616	Internal ECU monitoring detection reported error
523612	12	617	Internal ECU monitoring detection reported error
523612	12	618	Internal ECU monitoring detection reported error
523612	12	619	Internal ECU monitoring detection reported error
523612	12	620	Internal ECU monitoring detection reported error
523612	12	621	Internal ECU monitoring detection reported error
523612	12	623	Internal ECU monitoring detection reported error
523612	12	624	Internal ECU monitoring detection reported error
523612	12	625	Internal ECU monitoring detection reported error
523612	12	627	Internal ECU monitoring detection reported error
523612	12	628	Internal ECU monitoring detection reported error

SPN	FMI	KWP	Description
523612	12	637	Internal ECU monitoring detection reported error
523612	12	1170	Internal software error ECU
523612	14	973	Softwarereset CPU SWReset_0
523612	14	974	Softwarereset CPU SWReset_1
523612	14	975	Softwarereset CPU SWReset_2
523613	0	856	Maximum positive deviation of rail pressure exceeded (RailMeUn0)
523613	0	857	Maximum positive deviation of rail pressure in metering unit exceeded (RailMeUn1)
523613	0	858	Railsystem leakage detected (RailMeUn10)
523613	0	859	Maximum negative deviation of rail pressure in metering unit exceeded (RailMeUn2)
523613	0	860	Negative deviation of rail pressure second stage (RailMeUn22)
523613	0	862	Maximum rail pressure exceeded (RailMeUn4)
523613	1	861	Minimum rail pressure exceeded (RailMeUn3)
523613	2	864	Setpoint of metering unit in overrun mode not plausible
523615	3	594	Metering unit (Fuel-System); short circuit to battery highside
523615	3	596	Metering unit (Fuel-System); short circuit to battery low side
523615	4	595	Metering unit (Fuel-System); short circuit to ground high side
523615	4	597	Metering Unit (Fuel-System); short circuit to ground low side
523615	5	592	Metering unit (Fuel-System); open load
523615	12	593	Metering unit (Fuel-System); powerstage over temperature
523619	2	488	Physical range check high for exhaust gas temperature upstream (SCR-CAT)
523698	11	122	Shut off request from supervisory monitoring function

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SPN	FMI	KWP	Description
523717	12	125	Timeout Error of CAN-Transmit-Frame AmbCon; Weather environments
523718	3	1488	SCR mainrelay; short circuit to battery (only CV56B)
523718	4	1489	SCR mainrelay; short circuit to ground (only CV56B)
523718	5	1486	SCR mainrelay; open load (only CV56B)
523718	12	1487	SCR mainrelay; powerstage over temperature (only CV56B)
523766	9	281	Timeout Error of CAN-Receive-Frame Active TSC1AE
523767	9	282	Timeout Error of CAN-Receive-Frame Passive TSC1AE
523768	9	283	Timeout Error of CAN-Receive-Frame Active TSC1AR
523769	9	284	Timeout Error of CAN-Receive-Frame Passive TSC1AR
523770	9	285	Timeout Error of CAN-Receive-Frame Passive TSC1DE
523776	9	291	Timeout Error of CAN-Receive-Frame TSC1TE - active
523777	9	292	Passive Timeout Error of CAN-Receive-Frame TSC1TE; Setpoint
523778	9	293	Active Timeout Error of CAN-Receive-Frame TSC1TR
523779	9	294	Passive Timeout Error of CAN-Receive-Frame TSC1TR

SPN	FMI	SPN	Description
523788	12	299	Timeout Error of CAN-Transmit-Frame TrbCH; Status Wastegate
523793	9	202	Timeout Error of CAN-Receive-Frame UAA10; AGS sensor service message
523794	9	203	Timeout Error of CAN-Receive-Frame UAA11; AGS sensor data
523895	13	559	Check of missing injector adjustment value programming (IMA) injector 1 (in firing order)
523896	13	560	check of missing injector adjustment value programming (IMA) injector 2 (in firing order)
523897	13	561	check of missing injector adjustment value programming (IMA) injector 3 (in firing order)
523898	13	562	check of missing injector adjustment value programming (IMA) injector 4 (in firing order)
523910	6	1261	Air Pump; over current
523913	3	74	Sensor error glow plug control diagnostic line voltage; signal range check high
523913	4	75	Sensor error glow plug control diagnostic line voltage; signal range check low
523914	3	78	Glow plug control; short circuit to battery
523914	4	79	Glow plug control; short circuit to ground
523914	5	76	Glow plug control; open load
523914	5	1216	Glow plug control release line; short circuit error
523914	11	1217	Glow plug control; internal error
523914	12	77	Glow plug control; powerstage over temperature
523919	2	1378	Sensor air pump airpressure; plausibility error
523920	2	1379	Sensor exhaust gas back pressure burner; plausibility error

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SPN	FMI	KWP	Description
523922	7	1262	Burner Shut Off Valve; blocked closed
523922	7	1264	Burner Shut Off Valve; blocked closed
523929	0	109	Fuel Balance Control integrator injector 1 (in firing order); maximum value exceeded
523929	1	115	Fuel Balance Control integrator injector 1 (in firing order); minimum value exceeded
523930	0	110	Fuel Balance Control integrator injector 2 (in firing order); maximum value exceeded
523930	1	116	Fuel Balance Control integrator injector 2 (in firing order); minimum value exceeded
523931	0	111	Fuel Balance Control integrator injector 3 (in firing order); maximum value exceeded
523931	1	117	Fuel Balance Control integrator injector 3 (in firing order); minimum value exceeded
523932	0	112	Fuel Balance Control integrator injector 4 (in firing order); maximum value exceeded
523932	1	118	Fuel Balance Control integrator injector 4 (in firing order); minimum value exceeded
523935	12	168	Timeout Error of CAN-Transmit-Frame EEC3VOL1; Engine send messages
523936	12	169	Timeout Error of CAN-Transmit-Frame EEC3VOL2; Engine send messages
523946	0	1158	Zero fuel calibration injector 1 (in firing order); maximum value exceeded

SPN	FMI	KWP	Description
523946	1	1164	Zero fuel calibration injector 1 (in firing order); minimum value exceeded
523947	0	1159	Zero fuel calibration injector 2 (in firing order); maximum value exceeded
523947	1	1165	Zero fuel calibration injector 2 (in firing order); minimum value exceeded
523948	0	1160	Zero fuel calibration injector 3 (in firing order); maximum value exceeded
523948	1	1166	Zero fuel calibration injector 3 (in firing order); minimum value exceeded
523949	0	1161	Zero fuel calibration injector 4 (in firing order); maximum value exceeded
523949	1	1167	Zero fuel calibration injector 4 (in firing order); minimum value exceeded
523960	0	1011	Physical range check high for EGR cooler downstream temp.
523960	0	1458	High exhaust gas temperature EGR cooler downstream; warning threshold exceeded
523960	1	1012	Physical range check low for EGR cooler downstream temp.
523960	1	1459	High exhaust gas temperature EGR cooler downstream; shut off threshold exceeded
523980	14	1187	Bad quality of reduction agent detected
523981	11	918	Urea-tank without heating function (heating phase)
523982	0	360	Powerstage diagnosis disabled; high battery voltage
523982	1	361	Powerstage diagnosis disabled; low battery voltage

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SPN	FMI	KWP	Description
523988	3	1245	Charging lamp; short circuit to battery
523988	4	1246	Charging lamp; short circuit to ground
523988	5	1243	Charging lamp; open load
523988	12	1244	Charging lamp; over temp.
523998	4	1327	Injector cylinder bank 2 slave; short circuit
523999	12	1328	Injector powerstage output Slave defect
524014	1	1254	Air pressure glow plug flush line; below limit
524016	2	1259	Amount of air is not plausible to pump speed
524016	2	1260	Calculated amount of air is not plausible to HFM reading
524016	11	1258	HFM sensor; electrical fault
524021	11	1263	Burner fuel line pipe leak behind Shut Off Valve
524024	11	1302	Deviation of the exhaust gas temp. setpoint to actual value downstream (DOC) too high
524028	2	1431	CAN message PROEGRActr; plausibility error
524029	2	1432	Timeout Error of CAN-Receive-Frame ComEGRActr - exhaust gas recirculation positioner
524030	7	1440	EGR actuator; internal error
524031	13	1441	EGR actuator; calibration error

SPN	FMI	KWP	Description
524032	2	1442	EGR actuator; status message EGRCust is missing
524033	7	1443	EGR actuator; due to overload in Save Mode
524034	3	1438	Disc separator; short circuit to battery
524034	4	1439	Disc separator; short circuit to ground
524034	5	1436	Disc Separator; open load
524034	12	1437	Disc Separator; powerstage over temperature
524035	12	1341	Injector diagnostics; time out error in the SPI communication
524057	2	1505	Electric fuel pump; fuel pressure build up error
524097	9	1663	Timeout error of CAN-Transmit-Frame DPFBnAirPmpCtl
524098	9	1664	Timeout error of CAN-Transmit-Frame ComDPFBnPT
524099	9	1665	Timeout error of CAN-Transmit-Frame ComDPFC1
524100	9	1666	Timeout error of CAN-Transmit-Frame ComDPFHisDat
524101	9	1667	Timeout error of CAN-Transmit-Frame ComDPFTstMon

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SPN	FMI	KWP	Description
524102	9	1674	Timeout error of CAN-Receive-Frame ComRxDPFBrnAirPmpCtl
524103	9	1675	Timeout error of CAN-Receive-Frame ComRxDPFBrnAirPmp
524104	9	1676	Timeout error of CAN-Receive-Frame ComRxDPFCl
524105	9	1668	Timeout error of CAN-Transmit-Frame ComEGRMsFlw
524106	9	1677	Timeout error of CAN-Receive-Frame ComRxEGRMsfW1
524107	9	1678	Timeout error of CAN-Receive-Frame ComRxEGRMsfW2
524108	9	1669	Timeout error of CAN-Transmit-Frame ComEGRTVActr
524109	9	1679	Timeout error of CAN-Receive-Frame ComRxEGRTVActr
524110	9	1670	Timeout error of CAN-Transmit-Frame ComETVActr
524111	9	1680	Timeout error of CAN-Receive-Frame ComRxETVActr
524112	9	1671	Timeout ComlTVActr
524113	9	1681	Timeout error of CAN-Receive-Frame ComRxITVActr

SPN	FMI	KWP	Description
524114	9	1659	Timeout error of CAN-Transmit-Frame A1DOC
524115	9	1660	Timeout error of CAN-Transmit-Frame AT1S
524116	9	1661	Timeout error of CAN-Transmit-Frame SCR2
524117	9	1662	Timeout error of CAN-Transmit-Frame SCR3
524118	9	1672	Timeout error of CAN-Receive-Frame ComRxCM1
524119	9	1673	Timeout error of CAN-Receive-Frame ComRxCustSCR3
524120	9	1682	Timeout error of CAN-Receive-Frame ComRxSCRHtDiag
524121	9	1683	Timeout error of CAN-Receive-Frame ComRxTrbChActr
524122	9	1684	Timeout error of CAN-Receive-Frame ComRxUQSens
524123	9	1685	Timeout error of CAN-Receive-Frame ComSCRHtCtl
524124	9	1686	Timeout error of CAN-Receive-Frame ComTxAT1IMG
524125	9	1687	Timeout error of CAN-Receive-Frame ComTxTrbChActr